MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6081

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: ATR, fixed frequency, Fo = 5,640 MHz

ABSOLUTE RATINGS:

Parameter:	Transmitter po	Du	tp	Alt
Unit:	kw	---	μs	ft
Maximum:	---	0.001	1.0	10,000
Minimum:	10	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any (See note 3)

TEST CONDITIONS:

Parameter:	po	prr	tp	σ'
Unit:	kw	pps	μs	---
Test 1:	20	1,000	1.0	---
Test 2:	70(min)	1,000	1.0	1.03(max)
Test 3:	300±10	1,000	1.0	1.2(max)

Frequencies		
F	MHz	±
F1	5,640	---
F2	5,640	0.1%
F3	5,640	4%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	---					
4482	Tuning susceptance	---	---	Fo = F2	B/Yo	-0.06	+0.06	---
4486	Firing time	2	1	Fo = F3	t	---	10	sec
4488	Arc loss	4	---		La	---	0.8	dB
1027	Temperature cycling (nonoperating)	5	---		---	1	---	Cycle
1027	Temperature cycling life test	---	---	Group D	---	10	---	Cycles
	Quality conformance inspection, part 2							
4484	Normalized conductance	---	---	Fo = F2	G/Yo	---	0.1	---
	Quality conformance inspection, part 3							
---	Life-test provisions	---	3	Group D, Fo F3	t	500	---	hrs
---	Life-test end points	---	---					
4488	Arc loss	---	1	Fo F3	La	---	1.0	dB
4484	Normalized conductance	---	---	Fo F2	G/Yo	---	0.1	---
	Periodic-check tests							
4021	Degradation due to vibration	---	---	No voltages	---	---	---	---
4461	Loaded Q	---	---	Fo F1	QL	---	7.0	---
4474	High-level VSWR	---	2	Fo - F1	σ'	---	1.10	---

NOTES:

1. All tests listed under quality conformance inspection, part 1, shall have an AQL of 1.0 percent and an inspection level of II.
2. This test to be performed at the conclusion of the holding period.
3. The tube mount shall be designed to avoid interference from a toroidal mode over the operating range of the tube. For this purpose a shorting contact or capacitive shunt may be included in the mount centered in area "A" as shown on figure 1.
4. This test shall be performed at least 7 days after pumping and at least 24 hours after any previous discharge.
5. This test may be conducted during the holding period but not less than 24 hours before the arc-loss test.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

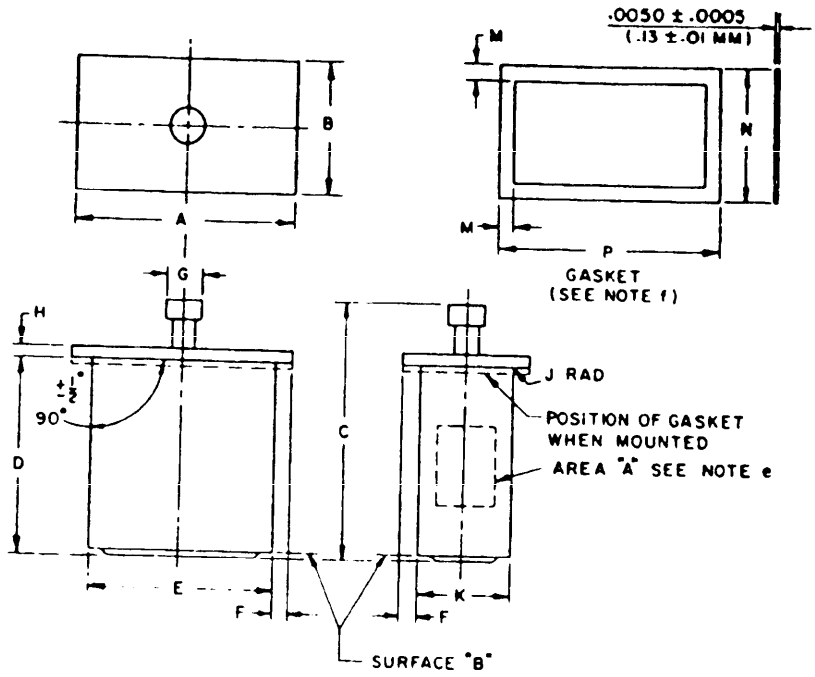
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-39)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			1
A	2.395 (60.83)	2.405 (61.09)	
B	1.395 (35.43)	1.405 (35.69)	
D	2.074 (52.68)	2.102 (53.39)	
E	1.985 (50.42)	2.015 (51.18)	
F	.182 (4.62)		c
K	.985 (25.02)	1.015 (25.78)	
P	2.395 (60.83)	2.405 (61.09)	
Quality conformance inspection, part 2			
C		2.750 (69.85)	
G		.375 (9.53)	
H	.117 (2.97)	.133 (3.38)	
J	(all sides)	.040 (1.02)	
M	.145 (3.68)	.155 (3.94)	
N	1.395 (35.43)	1.405 (35.69)	

NOTES:
a. The tubulation shall fall within a circle of .500 (12.70 mm) dia max located about the center of the flange.
b. Silver plate 100 msi min. or equivalent.
c. Applies at all edges of window face of tube at surface B.
d. Centerlines of window shall coincide with corresponding centerlines of the body within .015 (.38 mm) measured in plane of window.
e. Area A centrally located on barrel (see note 3).
f. Gasket is soft temper nickel or copper, two supplied with tube but not mounted.

FIGURE 1. Outline drawing of electron tube type 6081.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-2525

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